1. Compute the Walbright Fund's time-weighted rate of return.

Solution:

Because interim cash flows were made on 1 May, we must compute two interim total returns and then link them to obtain an annual return. Exhibit 15 lists the relevant market values on 1 January, 1 May, and 31 December, as well as the associated interim four-month (1 January to 1 May) and eight-month (1 May to 31 December) holding period returns.

Exhibit 15: Cash Flows for the Walbright Fund

1 January	Beginning portfolio value = USD100 million
1 May	Dividends received before additional investment = USD2 million
	Ending portfolio value = USD110 million
	Four-month holding period return: $R = \frac{USD2 + USD10}{USD100} = 12\%$
	New investment = USD20 million
	Beginning market value for last two-thirds of the year = USD132 million
31 December	Dividends received = USD2.64 million
	Ending portfolio value = USD140 million
	Eight-month holding period return: $R = \frac{USD2.64 + USD140 - USD132}{USD132} = 8.06\%$

Now we must geometrically link the four- and eight-month holding period returns to compute an annual return. We compute the time-weighted return as follows:

$$R_{TW} = 1.12 \times 1.0806 - 1 = 0.2103.$$

In this instance, we compute a time-weighted rate of return of 21.03 percent for one year. The four-month and eight-month intervals combine to equal one year. (Note: Taking the square root of the product 1.12 × 1.0806 would be appropriate only if 1.12 and 1.0806 each applied to one full year.)

2. Compute the Walbright Fund's money-weighted rate of return.

Solution:

To calculate the money-weighted return, we need to find the discount rate that sets the sum of the present value of cash inflows and outflows equal to zero. The initial market value of the fund and all additions to it are treated as cash outflows. (Think of them as expenditures.) Withdrawals, receipts, and the ending market value of the fund are counted as inflows. (The ending market value is the amount investors receive on liquidating the fund.) Because interim cash flows have occurred at four-month intervals, we must solve for the four-month internal rate of return. Exhibit 15 details the cash flows and their timing.

$$CF_0 = -100$$

$$CF_1 = -20$$

$$CF_2 = 0$$